

Threat Analysis & Risk Assessment

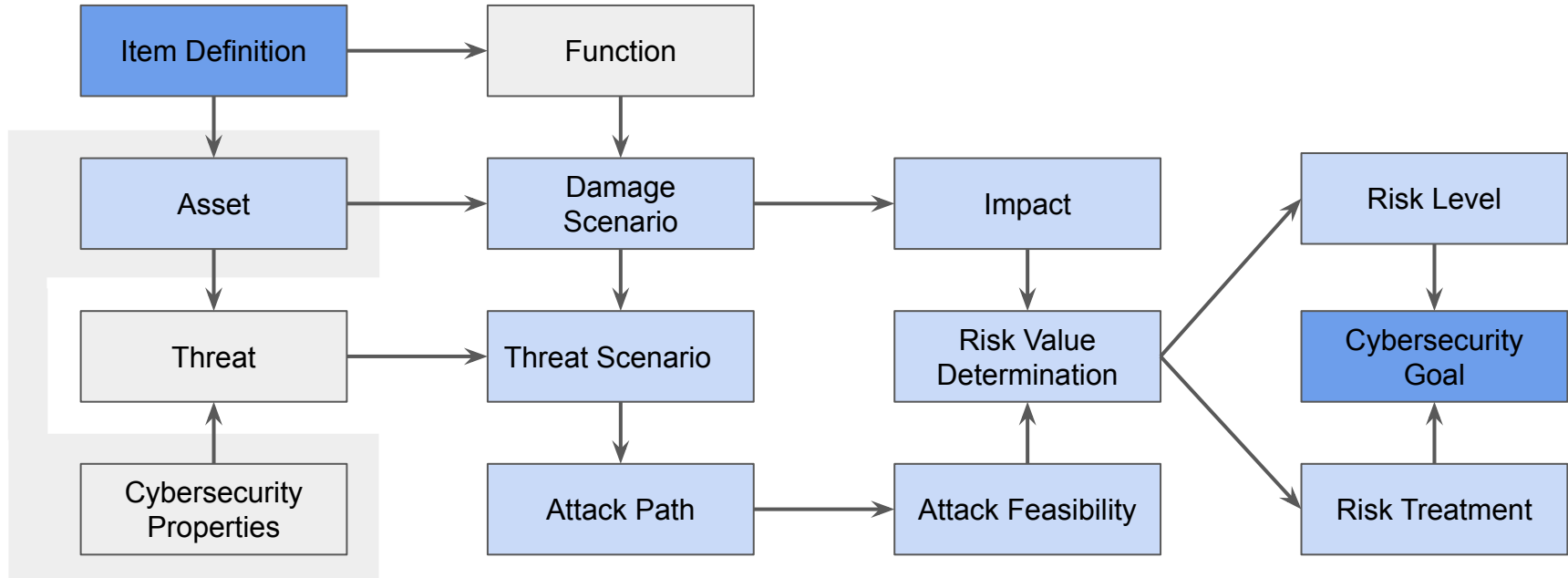
As per ISO 21434

General Notes - Vatsal Patel

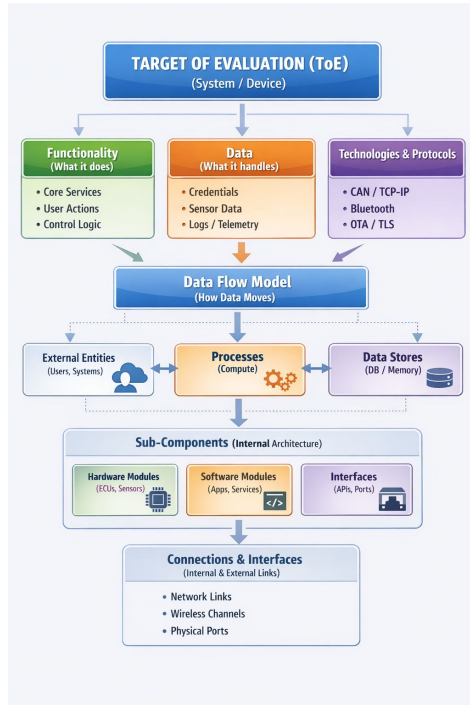
TARA Objective



Methodology



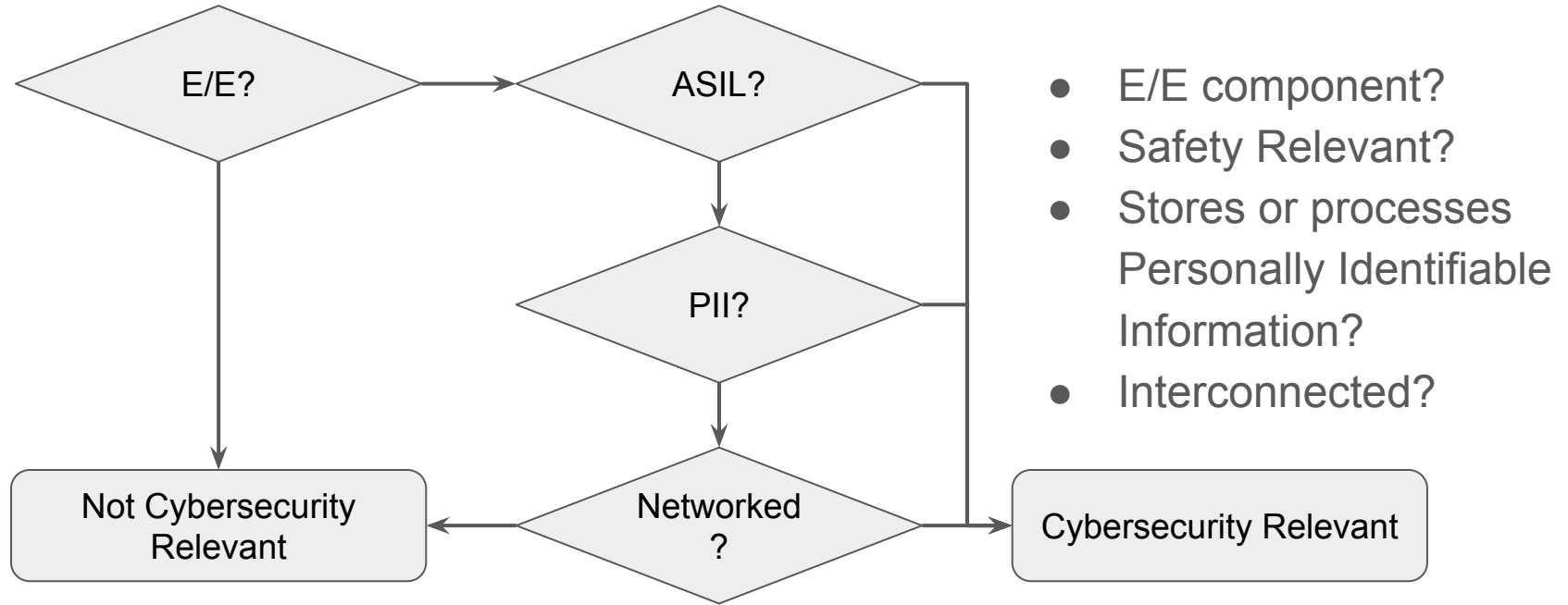
Modeling Item / Target of Evaluation (ToE)



*Image generated by providing written description to ChatGPT

- Functions
- Data
- Technologies
- Components and Connections
- Data Flow
- Diagram
- Mapping
 - Component - Function
 - Function - Data Flow

Cybersecurity Relevance



Item / Target of Evaluation (ToE) Definition [WP-09-01]

- All Product Information
- Main Function
- Architecture
- Operational Environment
- Scope

Asset Identification

Damage Scenario [WP-15-01]

- Relationship between functionality and adverse consequences.
- Description of damage/harm to the road user

Assets with Cybersecurity properties, [WP-15-02]

whose compromise leads to harm situation.

Threat Scenario Identification [WP-15-03]

Specific Asset

Compromises cybersecurity property

Cause of compromise

- Method, tools, attacker, attack surface

Methods

- Identification of abuse and misuse cases
- Framework - STRIDE

Potential Threat

Data Element	S	T	R	I	D	E
External Entity	X		X			
Data Flow		X		X	X	
Stored Data		X	X	X	X	
Data Process	X	X	X	X	X	X

Impact Rating [WP-15-04]

Impact Categories:

- Safety Annex F.2 (derived/reused from ISO 26262-3:2018, 6.4.3)
- Financial Annex F.3
- Operational Annex F.4
- Privacy Annex F.5

Degree of Impact:

- Severe, Major, Moderate, Negligible.

Note:

Impact with the highest criticality from any of these four category shall be analyzed, rest can be omitted.

Attack Paths [WP-15-05]

Attack Surface & Proximity

- Physical (e.g. Pins, Connectors...)
- Local (e.g. LIN, CAN, USB...)
- Adjacent (e.g. BT, WI-FI, Sensor...)
- Remote (e.g. Internet, Backed...)

Attack Feasibility Rating [WP-15-06]

Attack Vector:

- Physical, Local, Adjacent, Network

Attack Potential:

- Elapsed time, expertise, system knowledge, window of opportunity, Equipment.

CVSS 3.1 (or 4.0): Exploitability Metrics:

- Attack Vector, Attack Complexity, (Attack Requirement,) Privileges required, User Interaction

Risk -Value Determination [WP-15-07] & -Treatment [WP-15-08]

Impact Rating	Attack Feasibility Rating			
	Very low	Low	Medium	High
Severe	2	3	4	5
Major	1	2	3	4
Moderate	1	2	2	3
Negligible	1	1	1	1

Treatment

- Accept (Reatin)
- Transfer / Share
- Avoid
- Reduce / Mitigate

Cybersecurity Activity & Work products (as per ISO 21434)

Sub-clauses	Work Products (WP)
Concept Phase	
9.3 Item Definition	[WP-09-01] Item Definition
9.4 Cybersecurity Goals	[WP-09-02] TARA [WP-09-03] Cybersecurity Goal ...
Threat Analysis and Risk Assessment Methods	
15.3 Asset Identification	[WP-15-01] Damage Scenarios [WP-15-02] Assets with Cybersecurity Properties
15.4 Threat Scenario Identification	[WP-15-03] Threat Scenarios
15.5 Impact Rating	[WP-15-04] Impact ratings with associated Impact Categories
15.6 Attack Path Analysis	[WP-15-05] Attack Paths
15.7 Attack Feasibility Rating	[WP-15-06] Attack Feasibility Ratings
15.8 Risk Value Determination	[WP-15-07] Risk Values
15.9 Risk Treatment Decision	[WP-15-08] Risk Treatment Decisions